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$$\lim_{x\to\frac{1}{4}} \frac{4x^3+13x^2+\frac{5}{2}x-\frac{3}{2}}{3x^2+\frac{13}{4}x-1} = \frac{0}{0}$$

Horner:

$$\begin{array}{c|cccc} & 4 & 13 & \frac{5}{2} & \frac{-3}{2} \\ \frac{1}{4} & \downarrow & 1 & \frac{7}{2} & \frac{3}{2} \\ \hline & 4 & 14 & 6 & 0 \end{array}$$

$\to 4x^2+14x+6$

$$\begin{array}{c|ccc} & 3 & \frac{13}{4} & -1 \\ \frac{1}{4} & \downarrow & \frac{3}{4} & 1 \\ \hline & 3 & 4 & 0 \end{array}$$

$\to 3x+4$

$$\Rightarrow \lim_{x\to\frac{1}{4}} \frac{\left(x-\frac{1}{4}\right)(4x^2+14x+6)}{\left(x-\frac{1}{4}\right)(3x+4)} = \lim_{x\to\frac{1}{4}} \frac{4x^2+14x+6}{3x+4} = \frac{\frac{39}{4}}{\frac{19}{4}} = \frac{39}{19}$$

References